

STAD29 / STA 1007 assignment 4

Due Tuesday Feb 26 at 11:59pm on Quercus

Hand in the indicated questions. In preparation for the questions you hand in, it is worth your while to work through (or at least read through) the other questions as well.

Hand in your work on Quercus. If you did STAC32 last fall, it's the same procedure. A reminder is here: <https://www.uts.utoronto.ca/~butler/c32/quercus1.nb.html>

You are reminded that work handed in with your name on it must be *entirely your own work*. It is as if you have signed your name under it. If it was done wholly or partly by someone else, *you have committed an academic offence*, and you can expect to be asked to explain yourself. The same applies if you allow someone else to copy your work. The grader will be watching out for assignments that look suspiciously similar to each other (or to my solutions). Besides which, if you do not do your own assignments, you *will* do badly on the exams, because the struggle to figure things out for yourself is an important part of the learning process.

You will need this first:

```
library(tidyverse)
```

Hand in questions 2 and 4.

1. Work through, or read through, problems 19.1 and 19.2 of PASIAS.
2. One of the things your kidney does is to filter toxic substances out of the blood. For people whose kidneys do not work properly, they have to have treatments with a “dialysis machine” that does what the kidney is supposed to do. People who use dialysis machines often suffer weight gain between treatments due to fluid buildup. In our data, weight gain is classified into “mild”, “moderate” and “severe”. There are two types of dialysis treatment, called “short” and “long”. The outcome measure was the number of days that each patient spent in hospital during a year (because of the kidney failure or because of other problems that it caused).

Our aim in this question is to learn what effects, if any, the duration of treatment and the level of weight gain have on number of days in hospital. The data for a study are in <https://www.uts.utoronto.ca/~butler/d29/kidney.csv>.

- (a) (2 marks) Read in and display (some of) the data.
- (b) (3 marks) Draw a grouped boxplot that shows how the number of days in hospital, called `hospital_days`, depends on duration of treatment and weight gain.
- (c) (2 marks) Why does your plot suggest that a transformation of `hospital_days` would be a good idea? Explain briefly.
- (d) (4 marks) The original researchers decided to make a log transformation of `hospital_days` plus one (because some of the values of `hospital_days` were zero). This is in the column called `log_days` in the data frame. Find the mean `log_days` for each duration and weight gain, and use the resulting data frame to make an interaction plot.
- (e) (2 marks) What do you conclude from your interaction plot? Explain briefly.
- (f) (3 marks) Run an ANOVA predicting the transformed number of days in hospital from the weight gain and duration. Include an interaction. From the results, what is the most important conclusion?
- (g) (3 marks) Do you need to keep the interaction term? If you do, explain briefly why; if not, remove it and display the results for your new model. What do you conclude now, at $\alpha = 0.10$?

- (h) (3 marks) Run a suitable Tukey analysis. What can you say about the combination of weight gain and duration that is associated with the *smallest* mean number of days in hospital? Explain briefly. (Use $\alpha = 0.10$ again.)
3. Work through, or read through, problems 19.3 through 19.6 of PASIAS.
4. Nuclear power plants are often next to areas of water such as lakes or bays because they need water for cooling their condensers. This water is then discharged back into the lake or bay. Does that discharged water have a detrimental effect on the plant life in the water? Researchers at power plant located by a bay investigate three locations, called “stations” in the data set. Stations A and B are located close to the power plant’s discharge tubes, and station C is located further out in the bay. During one randomly chosen day in each of six months, a diver descends to the bottom at each of the stations and counts the number of blades of grass of a particularly important type found in a square-metre area. (The diver counts other things as well, but this is what is of interest to us.)
- The data are at <https://www.uts.utoronto.ca/~butler/d29/nuclear.csv>.
- The study has two aims:
- to determine whether stations A and B are similar or different in terms of the number of blades of grass of this type that they have.
 - to see whether station C, far from the power plant, has more blades of grass of this type than the (average of the) stations A and B, close to the power plant.
- (a) (2 marks) Explain briefly why contrasts would be a useful technique here.
- (b) (2 marks) Read in the data and explain briefly why you have a sensible number of observations.
- (c) (3 marks) Make a suitable plot of the blades of grass and the stations. What do you observe (two things), bearing in mind the aims of the study?
- (d) (2 marks) In your data frame, turn `station` into a **factor** (or create a new column that is a **factor** version of `station`). Save your new column in your original data frame. (The levels of the factor, and their order, will be exactly what you guess.)
- (e) (2 marks) Create two vectors with suitable names that reflect the desired contrasts that match with the study objectives.
- (f) (2 marks) Demonstrate that your two contrasts are orthogonal.
- (g) (2 marks) Turn these into contrasts for the column `station` that is in your data frame. (Two steps.)
- (h) (2 marks) Fit the analysis of variance using `lm`, and display the summary of the result.
- (i) (2 marks) What do we conclude from the analysis, bearing in mind the aims of the study?

Notes

¹I needed `MASS` for this, which I surreptitiously loaded first.

²If it were “loose” in your environment, instead of being in a data frame, it might work without the `aes`. This is what R people call “non-standard evaluation”.